

Cybernetic Security

Protection of endpoints

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Lecture Agenda

Antivirus, antimalware
Personal firewall
Unified endpoint management
Control of endpoints access to the network

Intro – Endpoints

- Endpoint = a client device, e.g. :
 - PC or laptop (OS: Windows, MacOS, Linux, ...),
 - Mobile device (OS: Android, iOS).
 - excluding servers, IoT devices etc.
 - NB: the above definition holds in this course only.
- Endpoint protection principle:
 - Control of all inputs:
 - Networks connections
 - wired – Ethernet,
 - wireless – WiFi, 4/5G, Bluetooth.
 - Local files (USB disks, internal storage).

Intro –Endpoint Risks

- Main endpoint risks:
 - Malicious software infection (virus, malware, trojan),
 - Remote (silent) connection (via e.g. a worm),
 - User deception into:
 - downloading a malicious app,
 - revealing sensitive information (login data),
 - ...

Malicious Code Detection Methods

- Signature-based:
 - Finding of known strings in executable code:
 - Not reliable for “stealth” virus.
- Heuristic approach:
 - Based on signature detection, with not exact match.
- Real-time detection:
 - Detecting suspicious activity of a code in OS, e.g.:
 - Writing OS core files,
 - Modification of system databases (like Registry, /etc directory, ...).
- User-privilege manipulation detection
 - Rootkit protection.

Protecting from Viruses & Malware

- Every software app that is executed can be potentially harmful.
 - The application is checked before execution:
 - for presence of known risks (e.g. virus "signatures") in the app code,
 - (optionally/limited) for being harmful for the OS (in real time):
 - e.g. accessing OS disk parts, communicating with network, scanning memory etc.
- More thorough check can be performed during computer idle periods.
- Also other data must be checked:
 - that can be converted into executable apps (e.g. message attachments).

Antivirus & Antimalware

- Local protection software:

- Searching for known patterns in incoming data:
 - In e-mail messages (incoming),
 - In web pages (being downloaded),
 - In stored files (on user request),
 - ...
 - With subsequent actions:
 - denying to display/save, sometimes with option to change the rule.
 - Search patterns (“virus database” MUST be updated in a short period (e.g. daily).
- Searching for suspicious (potentially risky) behavior:
 - Modification of system data,
 - Remote modification of locally stored files/databases,...
 - Subsequent action similar as above.

Antivirus & Antimalware

- Included in common OS:
 - Mainly desktop (Windows).
- It should be used on any device:
 - PCs, laptops:
 - including MacOS and Linux: used to be believed to be secure.
 - Mobile devices:
 - Preferably both Android and iOS
- Additional software features related to securing the device may be included in antivirus/antimalware, e.g.:
 - virus quarantine,
 - Analysis of suspicious files,
 - Sandbox testing,
 - ...

Other Client-side Protecting Software

- Registry protection:
 - for Windows,
 - other OS use different approach to store the config data.
- Cookie/Password managers,
- AdBlocker,
- ...
- Often integrated:
 - in a software package, or
 - in another software (e.g. a web browser).

Personal Firewall

- Local protection software:
 - Unlike network firewalls protecting many computers:
 - whole network or its segment.
- Firewall: set of RULES:
 - this is true for personal and network firewalls.
- By default, RULES block **ALL** incoming traffic (packets):
 - DENY ANY implicit rule,
 - except those explicitly allowed in advance (by PASS rules).
- They protect a device from:
 - Worms
 - APT,
 - ...

Personal Firewall

Rule List Example (Windows)

Group Policy Management Editor

File Action View Help

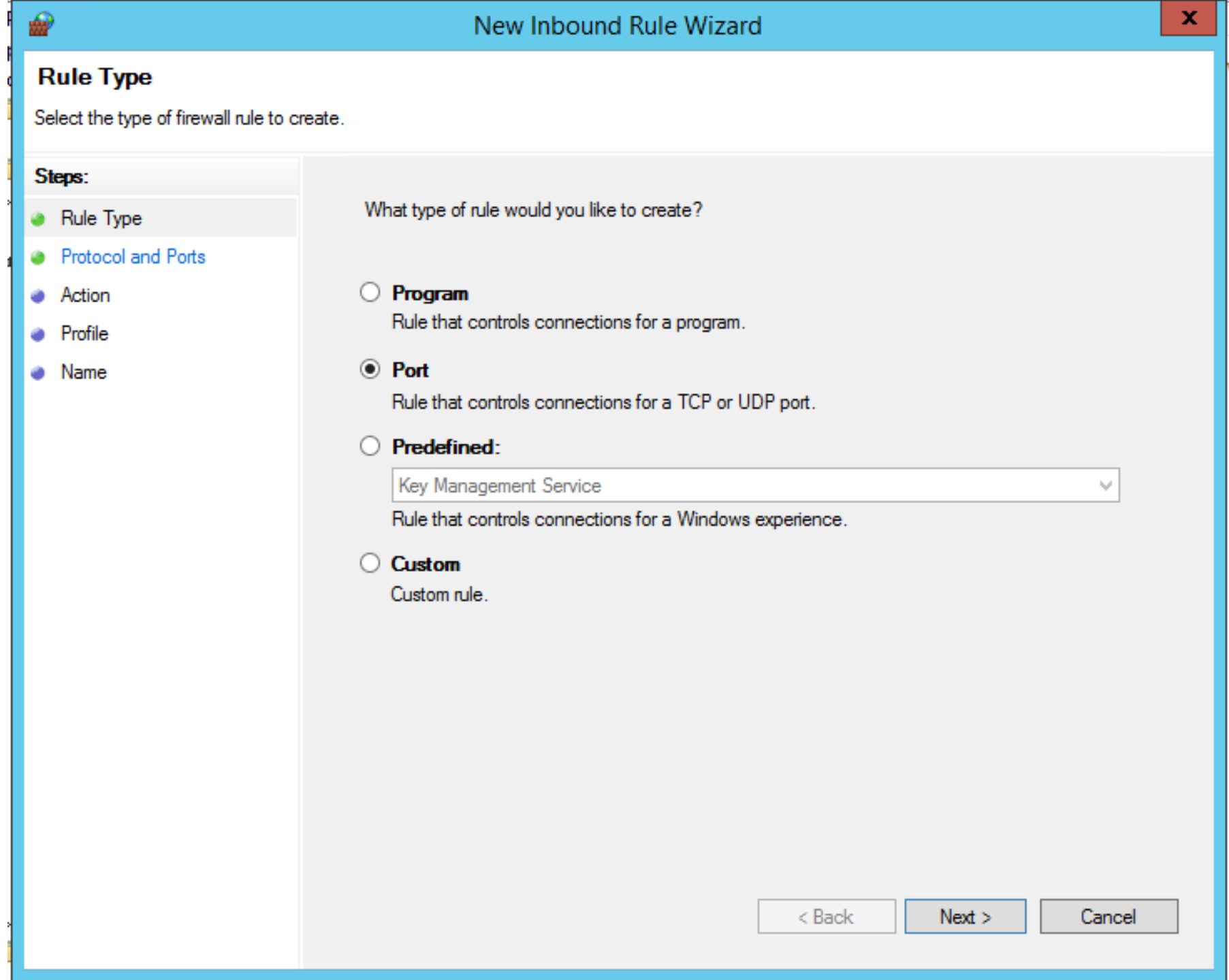
← → ↗ ↘ ? ▶

▲ Policies

- ▲ Software Settings
 - Software installation
- ▲ Windows Settings
 - ▶ Name Resolution Policy
 - Scripts (Startup/Shutdown)
- ▲ Security Settings
 - ▶ Account Policies
 - ▶ Local Policies
 - ▶ Event Log
 - ▶ Restricted Groups
 - ▶ System Services
 - ▶ Registry
 - ▶ File System
 - ▶ Wired Network (IEEE 802.3) Policies
 - ▲ Windows Firewall with Advanced Security
 - ▲ Windows Firewall with Advanced Security - Local
 - ▶ Inbound Rules
 - ▶ Outbound Rules
 - ▶ Connection Security Rules
 - Network List Manager Policies

Name	Group	Profile	Enabled	Action
✓ Windows Management Instrumentation ...	Windows Management Instr...	All	Yes	Allow
✓ Windows Management Instrumentation ...	Windows Management Instr...	All	Yes	Allow
✓ Windows Management Instrumentation ...	Windows Management Instr...	All	Yes	Allow
✓ Inbound Rule for Remote Shutdown (RP...	Remote Shutdown	All	Yes	Allow
✓ Inbound Rule for Remote Shutdown (TC...	Remote Shutdown	All	Yes	Allow
✓ Windows Remote Management (HTTP-In)	Windows Remote Manage...	Public	Yes	Allow
✓ Windows Remote Management (HTTP-In)	Windows Remote Manage...	Domai...	Yes	Allow
✓ File and Printer Sharing (LLMNR-UDP-In)	File and Printer Sharing	All	Yes	Allow
✓ File and Printer Sharing (Echo Request - I...	File and Printer Sharing	All	Yes	Allow
✓ File and Printer Sharing (Echo Request - I...	File and Printer Sharing	All	Yes	Allow
✓ File and Printer Sharing (Spooler Service -...	File and Printer Sharing	All	Yes	Allow
✓ File and Printer Sharing (Spooler Service -...	File and Printer Sharing	All	Yes	Allow
✓ File and Printer Sharing (NB-Datagram-In)	File and Printer Sharing	All	Yes	Allow
✓ File and Printer Sharing (NB-Name-In)	File and Printer Sharing	All	Yes	Allow
✓ File and Printer Sharing (SMB-In)	File and Printer Sharing	All	Yes	Allow
✓ File and Printer Sharing (NB-Session-In)	File and Printer Sharing	All	Yes	Allow
✓ AllowRDPIn		All	Yes	Allow

Personal Firewall Configuration Example (Windows): Rule types



The screenshot shows the 'New Inbound Rule Wizard' window in Windows. The title bar is blue with the text 'New Inbound Rule Wizard' and a close button. The main area is white. On the left, there is a 'Steps:' section with a list of steps: 'Rule Type' (selected with a green dot), 'Protocol and Ports' (green dot), 'Action' (purple dot), 'Profile' (purple dot), and 'Name' (purple dot). The main content area is titled 'Rule Type' and contains the instruction 'Select the type of firewall rule to create.' Below this, it asks 'What type of rule would you like to create?'. There are four radio button options: 'Program' (Rule that controls connections for a program.), 'Port' (selected, Rule that controls connections for a TCP or UDP port.), 'Predefined:' (Rule that controls connections for a Windows experience. with a dropdown menu showing 'Key Management Service'), and 'Custom' (Custom rule.). At the bottom right, there are three buttons: '< Back', 'Next >', and 'Cancel'.

New Inbound Rule Wizard

Rule Type

Select the type of firewall rule to create.

Steps:

- Rule Type
- Protocol and Ports
- Action
- Profile
- Name

What type of rule would you like to create?

☐ **Program**
Rule that controls connections for a program.

☒ **Port**
Rule that controls connections for a TCP or UDP port.

☐ **Predefined:**
Rule that controls connections for a Windows experience.
Key Management Service

☐ **Custom**
Custom rule.

< Back Next > Cancel

Unified Endpoint Management

- Evolution phase of device management through an organization.
 - Formerly Mobile Device Management:
 - Remote control of mobile devices,
 - Surveillance over protected data (corporate data),
- UEM: allows remote control of any device
 - Incl. PCs, tablets, ...
 - Common functions:
 - Provisioning (putting a device into service),
 - Software deployment,
 - App/event monitoring/tracking,
 - Updating,
 - Security configurations.

BYOD/Mobile Device Management

■ BYOD: Bring Your Own Device:

- An approach when an organization allows users to enter internal network from user-owned (mobile) device:
 - Exchanging for taking over a control over a device,
 - At least corporate data stored in the device.

■ Mobile Device Management (MDM):

- Software tool for managing mobile devices:
 - Necessary for BYOD devices
 - Features:
 - Mobile device tracking&Remote control:
 - Incl. optional deleting them when a device is lost.
 - Mobile device management (deployment, app management, data surveillance etc.),
 - Separation of private data and corporate data, app containerization,
 - Identity management (IAM),
 - Device security.

Control of Endpoints Access to Networks

■ Network Admission/Access Control:

- Software aimed to check meeting basic security requirements on the client:
 - For instance, a laptop/tablet reconnects to the network after a period:
 - Being connected somewhere else (thus potentially risky)
- Security requirements may include:
 - Up-to-date patches applied to OS,
 - Up-to-date antivirus/antimalware,
 - Detailed antivirus check passed less than e.g. 24 hrs before connecting to the network,
 - ...

Network Admission Control

- 2 possible approaches:

- Pre-admission:

- Logging-in is conditioned by security checks:
 - The delay may be significant (annoying for users).

- Post-admission:

- Check happen after logging-in:
 - Before completing the security check, the client permissions in the network are limited, e.g.:
 - Enforced re-authentication when accessing computers/server resources in other part of the network,
 - Limited connection to other devices in the network.

Network Admission Control

■ Client/server software:

- Client: usually executable with ability to:
 - Control OS starting process (primarily logging into network)
 - Protected against by-passing during OS start:
 - Must run anytime when the device is reconnected to the corporate network.
- Server:
 - The only network service available for clients before logging-in.
 - Provides security services like:
 - Recent patches for all supported OSs, including tools for checking of installed OSs and patches,
 - Downloadable antivirus/antimalware for supported Oss.
 - Can operate as authentication server (e.g. RADIUS):
 - After the client passes the check, the VLAN change (IEEE 802.1x) is performed.

Questions? Comments?

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Thank you for your attention

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